Note: Problems 1–3 are "fundamental" theorems for the real numbers. As such, you should take special care to *only* use the axioms on the "Number System Axioms" handout and results that we have proved in class.

Problem 1. Prove that for all $a \in \mathbb{R}$, we have $0 \cdot a = 0$.

Solution. Let $a \in \mathbb{R}$. We will show that $0a = 0$. By the additive identity axiom, we know that

$$0 + 0 = 0. \tag{1}$$

Using Theorem 7.1 from class, we can multiply both sides of Equation (1) by a, and get

$$(0 + 0)a = 0a. \tag{2}$$

Using the distribution axiom, we have $(0 + 0)a = 0a + 0a$, so using the substitution axiom on the left hand side of Equation (2), we have

$$0a + 0a = 0a.$$

By the additive identity axiom and commutativity, $0a = 0a + 0 = 0 + 0a$ so by substitution, we have

$$0a + 0a = 0 + 0a$$

Then, by the Cancellation Theorem (Theorem 7.2 from class), we see that

$$0a = 0.$$

Which was what we wanted. This completes our proof. □

Problem 2. Prove that for all $a \in \mathbb{R}$, we have $-a = (-1)a$.

Solution. Let $a \in \mathbb{R}$. Since -1 is the additive inverse of 1, by the additive inverse axiom, we know that

$$1 + (-1) = 0. \tag{3}$$

Multiplying both sides of Equation (3) by a, using Theorem 7.1, we get

$$(1 + (-1))a = 0a. \tag{4}$$

By Problem 1, we know that $0a = 0$. By the distributivity axiom, we know that $(1 + (-1))a = 1a + (-1)a$. So by two substitutions into Equation (4), we have

$$1a + (-1)a = 0.$$

Since 1 is the multiplicative identity, $1a = a$. So by another substitution, we have

$$a + (-1)a = 0. \tag{5}$$

Adding $-a$ (the additive inverse of a) to both sides of Equation (5) using Theorem 7.1 yields

$$(a + (-1)a) + (-a) = 0 + (-a).$$

Using commutativity on the left-hand side and substituting yields

$$-a + (a + (-1)a) = 0 + (-a).$$

By associativity, $-a + (a + (-1)a) = (-a + a) + (-1)a$. By the additive identity axiom, we have $-a + a = 0$ so by transitivity, $-a + (a + (-1)a) = 0 + (-1)a$. Substituting this into the left-hand side of the previous equation yields

$$0 + (-1)a = 0 + (-a). \tag{6}$$

Now using commutativity and the fact that 0 is the additive identity, we have $0 + (-1)a = (-1)a$ and $0 + (-a) = -a$. So two substitutions into Equation (6) yields

$$(-1)a = -a.$$

which is precisely what we wanted to show. This completes our proof. □

Problem 3. Prove that for all $a, b \in \mathbb{R}$, we have $(-a)(-b) = ab$.

Solution. Let $a, b \in \mathbb{R}$. By Problem 2, we know that $-b = (-1)b$. Therefore, by making a substitutions, we have

$$(-a)(-b) = (-a)((-1)b). \tag{7}$$

Now, by associativity $(-a)((-1)b) = ((-a)(-1))b$. By commutativity, $(-a)(-1) = (-1)(-a)$ so by substituting into the previous equality, we have

$$(-a)((-1)b) = ((-1)(-a))b.$$

Substituting this into the right-hand side of Equation (7) yields

$$(-a)(-b) = ((-1)(-a))b.$$

Again using Problem 2 of this assignment, we know that $-a = (-1)a$, and substitution yields

$$(-a)(-b) = (-1)((-1)a))b.$$

Now by associativity, we know $(-1)((-1)a) = ((-1)(-1))a$ and by a theorem from class, $(-1)(-1) = 1$. So we can make a substitution to conclude that $(-1)((-1)a) = 1a$. Substituting this into the previous displayed equation yields

$$(-a)(-b) = (1a)b.$$

Since 1 is the multiplicative identity, $1a = a$ and substituting yields

$$(-a)(-b) = ab$$

as desired. □

Problem 4. For each of the following statements, determine whether it is true or false. If it is true, write a *complete and formal* proof. If it is false, provide a counterexample (this should also be written in the same style).

1. For all $a, b, c \in \mathbb{Z}$ with $a \neq 0$, if $a \mid b$, then $a \mid bc$.
2. For all $a, b, c \in \mathbb{Z}$ with $a \neq 0$, if $a \mid (b - 1)$ and $a \mid (c - 1)$, then $a \mid (bc - 1)$.
3. For all $a, b, c \in \mathbb{Z}$ with $a \neq 0$, if $a \mid bc$, then $a \mid b$ or $a \mid c$.

Solution.

1. Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$. We will show that if $a \mid b$, then $a \mid bc$. (Recall that $x|y$ is shorthand for "x divides y.")

   If $a \mid b$, this means that $b = an$ for some $n \in \mathbb{Z}$. Therefore $bc = anc = a(nc)$. Since $nc \in \mathbb{Z}$ by the closure of $\mathbb{Z}$, this shows that $a \mid bc$. This completes the proof.

2. Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$. We will show that if $a \mid (b-1)$ and $a \mid (c-1)$, then $a \mid (bc-1)$.

   Assume that $a \mid (b-1)$ and $a \mid (c-1)$. Therefore there exist integers m, n such that

   $$b - 1 = am$$
   $$c - 1 = an.$$

   Solving for b and c, we see that

   $$b = am + 1$$
   $$c = an + 1.$$

   Notice that

   $$\begin{aligned} bc - 1 &= (am-1)(an-1) - 1 \\ &= a^2mn - am - an \\ &= a(amn - m - n). \end{aligned}$$

   Since $(amn - m - n) \in \mathbb{Z}$ (by the closure of the integers), this means that a divides $bc - 1$. □

3. We will show that the above statement is false by providing a counterexample. Let $a = 4$ and $b = c = 2$. Then $a = 4$ and $bc = 4$ and since $4 = 4 \cdot 1$, therefore $a \mid bc$.

   However, we know that $4 \nmid 2$ since if $2 = 4 \cdot k$, multiplying by 2^{-1} we obtain $1 = 2k$, which is no possible for any integer k since 1 is not even. Hence, in this case, $a \nmid b$ and $a \nmid c$. Hence, we have found a counterexample.

   (Another note: In some sense, this statement is "close" to being correct. For example, if $a = 2$ or $a = 3$, then the statement is true. Can you guess what additional conditions we'd need to add to make this statement true?)

Problem 5. Either prove the following statement is true or provide a counterexample to show that it is false:

For every $a, b \in \mathbb{Z}$, if $ab \equiv 0 \pmod 6$, then $a \equiv 0 \pmod 6$ or $b \equiv 0 \pmod 6$.

Solution. This statement is false. Let $a = 2$ and $b = 3$. Then $ab = 6$, so $ab \equiv 0 \pmod 6$, but $a \not\equiv 0 \pmod 6$ and $b \not\equiv 0 \pmod 6$. □

Problem 6. Let n be a positive integer. Prove the following statements about congruence modulo n. (Together, these prove that congruence modulo n is an *equivalence relation*.)

1. For all $a \in \mathbb{Z}$, $a \equiv a \pmod n$. (*Reflexivity*)
2. For all $a, b \in \mathbb{Z}$, if $a \equiv b \pmod n$ then $b \equiv a \pmod n$. (*Symmetry*)
3. For all $a, b, c \in \mathbb{Z}$, if $a \equiv b \pmod n$ and $b \equiv c \pmod n$, then $a \equiv c \pmod n$. (*Transitivity*)

Solution.

1. Let $a \in \mathbb{Z}$ and $n \in \mathbb{N}$. Notice that

   $$a - a = 0 \cdot n,$$

   so $a \equiv a \pmod n$.

2. Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{N}$. Assume that $a \equiv b \pmod{n}$. Therefore there is a $k \in \mathbb{Z}$ such that

$$a - b = kn. \tag{8}$$

Multiplying Equation (8) by -1, we see that

$$b - a = (-k)n.$$

Since $-k \in \mathbb{Z}$, this shows that $b \equiv a \pmod{n}$.

3. Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{N}$. Assume that $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. We will show that $a \equiv c \pmod{n}$. By the definition of congruence mod n, we know there exist integers k, ℓ such that

$$a - b = kn$$
$$b - c = \ell n.$$

Therefore

$$\begin{aligned} a - c = a - b + b - c &= (a - b) + (b - c) \\ &= kn + \ell n = (k + \ell)n. \end{aligned}$$

Therefore $a \equiv c \pmod{n}$. This completes the proof. □

Problem 7. Complete Exercise 19 parts (a), (b), and (c) on pages 100 and 101 of the textbook (*Mathematical Reasoning* by Sundstrom).

Solution.

(a) The proof has lots of issues. The writer doesn't describe any structure to the proof, and they don't give us basic information that we need to understand the proof (e.g., what are m and n?). It's a real mess. Here's the proposition and a better written proof.

Proposition. If m is an even integer, then $(5m + 4)$ is an even integer.

Proof. Let $m \in \mathbb{Z}$ be even. By definition of even, this means that $m = 2k$ for some $k \in \mathbb{Z}$. Therefore

$$5m + 4 = 5(2k) + 4 = 2 \cdot 5k + 2 \cdot 2 = 2(5k + 2).$$

Since $5k + 2 \in \mathbb{Z}$, this says precisely that $(5m + 4)$ is an even integer. This completes the proof. □

(b) This proposition is also true, but the proof is organized poorly. It starts with what we want to show and it works up to what we know. **This is a good way to figure out *how* to prove something, but it is *not* the correct format for a proof.** Instead, we should start with what we know and deduce the desired result. Here's the proposition and a better-organized proof.

Proposition. For all real numbers x and y, if $x \neq y$, $x > 0$, and $y > 0$, then $\frac{x}{y} + \frac{y}{x} > 2$.

Proof. Let x and y be positive real numbers such that $x \neq y$. We will show that

$$\frac{x}{y} + \frac{y}{x} > 2.$$

Since $x \neq y$, we know that $x - y \neq 0$, and therefore $(x-y)^2 > 0$. Distributing this multiplication, we see that

$$\begin{aligned} x^2 - 2xy + y^2 &> 0 \\ x^2 + y^2 &> 2xy. \end{aligned}$$

Since x and y are positive, so is xy and thus $\frac{1}{xy}$ is well-defined. Thus multiplying the above equation by $\frac{1}{xy}$ gives us

$$(x^2 + y^2) \cdot \frac{1}{xy} > 2xy \cdot \frac{1}{xy}$$
$$\frac{x}{y} + \frac{y}{x} > 2.$$

This is the desired result, which completes the proof. □

(c) Here's the problem: It's not clear that writing $k = mn$ is useful—what are m and n, anyway? They could be $m = 1, n = k$. Even if we know that $bc = mna$, there's no way that we can conclude that $b = ma$ or $c = na$.

As a concrete counterexample, consider $a = 10$, $b = 2$, and $c = 5$. Here $k = 1$, so $m, n = \pm 1$. But we cannot conclude that $b = 1 \cdot a$ or $c = 1 \cdot a$. □

Problem 8. Do not turn these in, but I recommend completing the progress checks in sections 3.1 and 3.2 in your textbook.